

TEAM NOEFFORT • DATA MINING

Early Warning System.

for ICU Sepsis Prediction

0112230367 – ABDULLAH AL NOMAN

0112230889 – MAHATHIR MOHAMMAD

0112230362 – RAKIBUL HASSAN

0112230117 – MD. ZAWAD AL MAHIR

Agenda.

SIX TOPICS · ONE MISSION

01

Key facts
about sepsis.

BACKGROUND

02

Problem &
motivation.

WHAT & WHY

03

Dataset &
pipeline.

DATA + WORKFLOW

04

Solution
approach.

HOW

05

Benchmarks &
closing.

EVALUATION

WHY EARLY DETECTION MATTERS

- Sepsis is a dangerous condition where the immune system overreacts to an infection and damages its own organs.
- It can lead to kidney failure, liver damage, heart problems, and even death.
- Intensive Care Unit (ICU) patients generate large amounts of medical data every hour — doctors may miss small changes.
- The goal is to predict sepsis before it becomes clinically obvious.
- Early detection can save lives and reduce ICU mortality rates.

The core challenge

EVERY HOUR COUNTS IN THE ICU

PROBLEM STATEMENT

What are we building?

INPUT → MODEL → OUTPUT

PROPOSAL 01

SEPSIS PREDICTION

PROBLEM DEFINITION

Predict sepsis hours before clinical onset.

We are building a machine learning model that analyzes ICU patient vital signs and lab results in real time to predict sepsis 4–6 hours before it becomes clinically apparent.

The model outputs a sepsis risk score (0–100%) and a binary classification — **no sepsis** or **sepsis likely** — giving clinicians a critical early warning.

INPUTS

8 vital + lab
features

OUTPUT

Risk score + binary

LEAD WINDOW

4–6 hours ahead

THREE REASONS WHY THIS MATTERS

- **Why important?** Sepsis is a leading cause of ICU mortality — early detection reduces death rates and saves thousands of lives every year.
- **Why not covered by state-of-the-art (SOTA)?** Current clinical tools — Sequential Organ Failure Assessment (SOFA), quick SOFA (qSOFA), Systemic Inflammatory Response Syndrome (SIRS) — are reactive. They detect sepsis only after organ dysfunction has already begun.
- **Why not tweak SOTA?** Existing machine learning (ML) solutions are tailored to specific hospital systems and lack real-time time-series modeling. Simple tuning cannot fix the class imbalance or the temporal nature of the data.

The gap we address

REACTIVE → PREDICTIVE

MOTIVATION

Heart Rate →

Blood Pressure →

Temperature →

Respiratory Rate →

SpO₂ →

Lab Results →

Age & Gender →

Sepsis Risk → 0–100%

VITALS + LABS

BINARY + PROBABILITY

THE PROBLEM

Predict sepsis before it is obvious.

Build a model that predicts whether a patient is likely to develop sepsis in the next few hours — using heart rate, blood pressure, temperature, respiratory rate, SpO₂, lab results, age, and gender.

The output is a binary classification (no sepsis / sepsis likely) or a probability score such as **sepsis risk = 91%**.

The data mining pipeline: clean missing values → extract time-series trends → select key features → balance classes with Synthetic Minority Over-sampling Technique (SMOTE) → train Random Forest + Gradient Boosting → explain predictions with SHapley Additive exPlanations (SHAP).

LEARNING TYPE

Supervised
learning

DATASET

PhysioNet CinC
2019

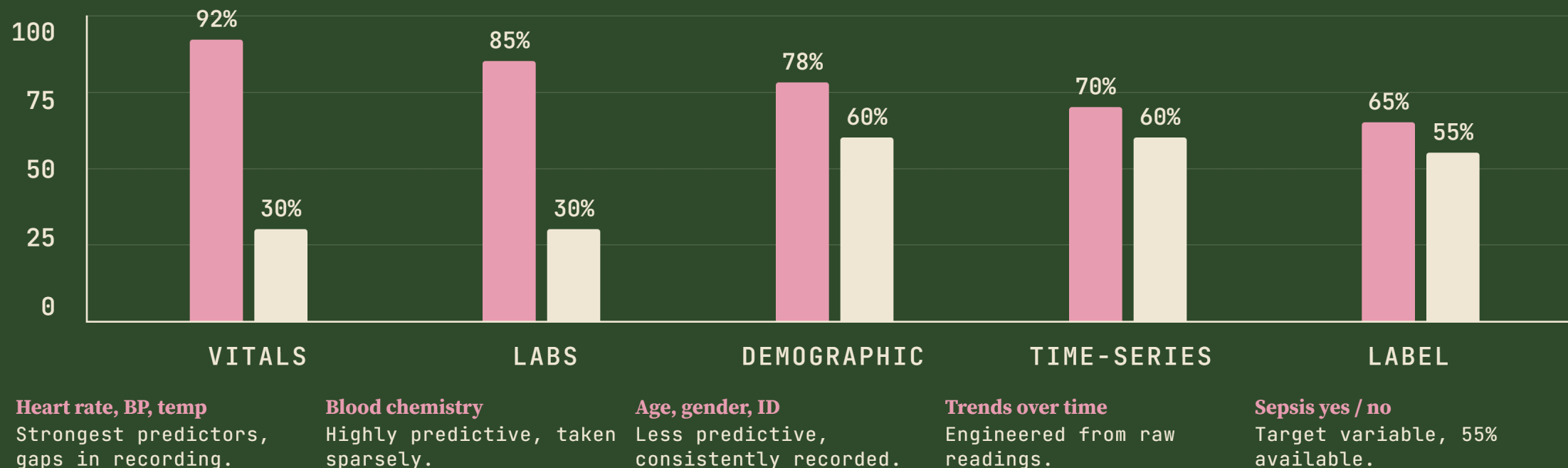
PATIENTS

40,000+ ICU

BY THE NUMBERS

Dataset features, from vitals to labels.

CLINICAL METADATA



SOURCE • PHYSIONET/CINC CHALLENGE 2019

40,374 PATIENTS • 8 FEATURE GROUPS

From raw data to explainable predictions.

Stage 01

Clean.

Remove missing values, incorrect entries, and duplicates from the dataset.

DATA

CLEANING

Stage 02

Engineer.

Extract time-series trends instead of single measurements — capture the trajectory.

FEATURES

TRENDS

Stage 03

Select & train.

Select key features, balance classes with SMOTE, then train Random Forest and Gradient Boosting ensemble models.

MODEL RF + XGBOOST

Stage 04

Explain.

Use SHAP to explain why a prediction was made — build trust with clinicians.

SHAP EXPLAINABILITY

THREE PILLARS OF OUR SOLUTION

- **Empirical study.** Train and evaluate ensemble models (Random Forest, XGBoost) on the PhysioNet 2019 dataset with rigorous cross-validation and SHAP explainability.
- **Simulation.** Simulate real-time ICU monitoring by feeding time-series windows into the trained model and measuring lead time before clinical confirmation.
- **Theoretical analysis.** Analyze feature importance distributions, model confidence calibration, and the impact of SMOTE on class boundary learning.

Hybrid approach

THEORY + SIMULATION + DATA

SOLUTION

How we measure success

AUROC · SENSITIVITY · LEAD TIME

BENCHMARKS

EVALUATION

BENCHMARK CASES

Rigorous comparison against baselines.

Dataset benchmark. PhysioNet/CinC Challenge 2019 — 40,374 ICU patients with hourly vitals and sepsis labels. The standard benchmark for sepsis prediction research.

Model comparison. Evaluate against SOFA score baseline, logistic regression, and published state-of-the-art results on the same dataset.

Metrics. Area Under the Receiver Operating Characteristic curve (AUROC), sensitivity/specificity, precision-recall curve, and lead time before clinical onset (hours ahead of SOFA).

BASELINE	PRIMARY METRIC	TARGET
SOFA + Logistic Regression	AUROC	≥ 4 hours lead time

WHAT IT ADDS UP TO

KEY METRICS

Scale, scope, and the hardest parts.

PATIENTS

40K+

ICU patients from the PhysioNet/CinC Challenge 2019 — one of the largest publicly available sepsis datasets.

FEATURES

8 categories

Vital signs, lab results, demographic data, and time-series trends engineered from raw measurements.

CHALLENGES

3 key

Missing values across time-series vitals, severe class imbalance (few sepsis positives), and high-dimensional temporal data — all addressed by SMOTE and feature engineering.

Predict. Explain. Save.

REAL IMPACT

A large, real-world healthcare dataset with high life-saving potential — early sepsis detection transforms ICU outcomes.

PROVEN APPROACH

Supervised learning with ensemble models, time-series features, SMOTE for imbalance, and SHAP for clinician trust.

BUILT TO EXPLAIN

Every prediction comes with a reason — SHAP explanations help doctors act on model output with confidence.

TEAM NOEFFORT • EARLY WARNING SYSTEM

Thank you.

Team NoEffort

0112230367 – Abdullah Al Noman

0112230362 – Rakibul Hassan

0112230889 – Mahathir Mohammad

0112230117 – Md. Zawad Al Mahir

QUESTIONS & DISCUSSION